

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

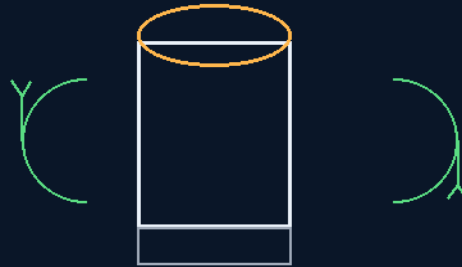
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 167

THE QUINN THE UNIVERSAL PLUNGER CONTAINER

Displace, never pour
one jar, every rack, zero disposable



the jar comes back

10L/10KG · SPIN LID · SEAL DOCK

ONE STANDARD FLEET-WIDE

Logistics system — v1.0 in force

GP-IM-167

Revision v1.0 — 23 August 2026

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VETTED BATCH 20 — PASS

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 167

PHASE 167: THE QUINN THE UNIVERSAL PLUNGER CONTAINER — CLOSED-LOOP SUPPLY STANDARD — STATUS: CURRENT — PASS; GRAVITY-INDEPENDENT CONTAINER SYSTEM. One container for everything the fleet carries, and it empties itself in any gravity. Phase 167 derives the fleet's 10 litre / 10 kilo standard container from the Phase 162 basket: segmented spin-close lid, rechargeable carry handle, seal-lock dock with plunger-assisted complete emptying — the plunger abolishes gravity-dependent pouring, so the last gram leaves the container in zero-G exactly as in 1G. The loop is closed by design: cleaning and trader refill with zero disposable stage — nothing aboard this container is ever thrown away. The audit's own note rides in §1, affirmed: the plunger abolishes gravity-dependent pouring. The vet passed the gravity-independent container system clean — the batch's final green of this twenty.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-167, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the container law as seated (contents line, the bodies — the container, the dock and empty cycle — the physics note and provenance, verbatim) — §2 why the plunger abolishes pouring (graded) — §3 the system in the container — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 167: **The Universal Plunger Container — Closed-Loop Supply Standard.** 10 litre/10 kilo standard container derived from the Phase 162 basket: segmented spin-close lid, rechargeable carry handle, seal-lock dock with plunger-assisted complete emptying in any gravity, cleaning and trader refill loop with zero disposable stage.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

Live instruction, 29 July 2026. The first child of Phase 162: the laundry basket's plunger lid, promoted from appliance to supply chain. Archer's framing opens the phase: *'"inventions create more inventions in a 3d direction"* — one mechanism, growing sideways into logistics, trade, and planetary resupply at once.

THE CONTAINER (VERBATIM)

- **The Standard:** any size possible — the fleet standard is 10 litre / 10 kilo. Take the Phase 162 laundry basket architecture, remove the mesh insert, and replace it with a segmented, spin-close lid and a rechargeable carry handle — the powered part of the unit.
- **What Fills It:** rice, mineral samples, any kind of food, nitropril (Phase 88's atmospheric compound) — filled by people on the ground or onboard, and sent to the ship. One container for every dry good the fleet moves. At stores scale the same architecture scales up: the multi-hundred-kilo ship silos are cylindrical too, with spring plungers keeping contents against the feed (see Phase 168).

THE DOCK & EMPTY CYCLE (VERBATIM)

- **Invert & Lock:** the container inverts onto a seal-lock storage unit — galley shelf, sample rack, or stores bay, all built to the same mouth.
- **Press, Open, Plunge:** press the button — the door opens; the plunger action ensures the container empties completely. Retract the plunger, press the button, the doors close.
- **Clean & Return:** the empty container goes to the cleaning unit, then back on the shelf. Empty units go down to the traders; small units come back filled — coffee in, the coffee machine refilled. And so on, forever — a closed loop with no disposable stage.

PHYSICS NOTE — PHASE 167 (KIMI AUDIT: AFFIRMED — THE PLUNGER ABOLISHES GRAVITY-DEPENDENT POURING) (VERBATIM)

- **The core problem is invisible on Earth and fatal in space:** granular goods pour because of gravity. Rice, flour, mineral samples — in zero-G none of them 'empty'; they float, drift, and contaminate. A plunger that sweeps the full interior makes emptying a mechanical certainty in any gravity frame — the same argument that made Phase 162's air-swept spin replace the second bowl. One mechanism, two domains: that is the 3D direction stated as engineering.
- **Standardization is the multiplier.** Earth's container revolution ran on one box and one twist-lock corner casting — every crane, truck, ship, and port built to the same interface. Phase 167 is the TEU

container at galley scale: one mouth geometry means the storage dock, the cleaning unit, the coffee machine, and the trader's shelf are all the same fitting, and any container serves any station. The 10L/10kg standard is human-factored — one person, one hand, full load.

- **The closed loop is mass-economics, not tidiness.** A century-ship cannot throw containers away — every gram of packaging launched is a gram of propellant spent forever. A container that cycles fill → ship → dock → plunge → clean → shelf → trader → refill has a consumables budget of zero; only its contents move. The nitropril cross-reference matters here: the same loop that carries rice carries the atmosphere program's feedstock, so logistics and life support share one standard.
- **Mineral samples complete the excursion chain.** Phases 165/166 equipped the crew on the surface; Phase 167 gives what they gather a ride home — samples sealed at the plunger mouth, docked at the lab rack, emptied without a glove ever touching dust in the air. Quarantine doctrine (Phase 12) plugs into the same mouth: the seal-lock dock is the agar-plated barrier's civilian cousin.

ORIGIN OF THOUGHT — PHASE 167 (VERBATIM)

Archer, 29 July 2026: "ok though we have spent most of the time collating and repairing you are about to see how we did 150 in 7 days. inventions create more inventions in a 3d direction. the newest one is our new container system, any size but we will do 10 litre/kilo. our laundry basket remove the mesh insert replace with a segmented spin close lid rechargeable carry handle. the people on the ground or onboard fill it with rice or mineral samples or any kind of foods some nitropril, goes to the ship. inverted on a seal lock storage units, press button door opens, plunger action ensures it empty. retract plunger, press button close doors, off to cleaning unit back on shelf. empty units taken down to the traders, small units, coffee in coffee machine refilled. etc etc forever."

Why it matters, in plain English: every supply chain in history has been built around gravity doing the last metre of work — the pour, the shake, the tap on the counter. Out here, that metre doesn't exist. This phase replaces the pour with the plunger and, in the same stroke, replaces the throwaway package with a permanent one: a container that is filled on a planet, flown to a ship, emptied by mechanism, washed, shelved, traded, and filled again — a loop with no end and no rubbish. And it was not designed from a whiteboard; it grew out of the laundry basket on an ordinary afternoon, which is exactly what '150 in 7 days' looked like the first time: each invention handing the next one its working parts.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE PLUNGER, AFFIRMED (the phase's own physics note): pouring needs gravity; a plunger needs none — positive displacement drives the contents to the dock regardless of orientation or acceleration. Complete emptying in any gravity is a mechanism claim, and the mechanism is correct.

THE STANDARD, AFFIRMED AS LOGISTICS: one size — 10 litres, 10 kilos — means every rack, dock, handle and inventory line in the fleet speaks one container. Standardization is the invisible machine; the phase names it the point.

THE LOOP, AFFIRMED: zero disposable stage — clean, refill, trade — closes the supply ring the file's whole closed-loop doctrine demands. The container is the galaxy's jar, and the jar comes back.

§3 — THE SYSTEM IN THE CONTAINER

- THE BODY: the 10 litre / 10 kilo standard, derived from the Phase 162 basket.
- THE LID: segmented spin-close.
- THE HANDLE: rechargeable carry handle.
- THE DOCK: seal-lock with plunger-assisted complete emptying — any gravity.
- THE LOOP: cleaning and trader refill — zero disposable stage.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE PLUNGER DOCTRINE (the phase's own note, affirmed): abolish gravity-dependent pouring — displace, never pour.
- THE ONE-STANDARD DOCTRINE: one container, every rack — logistics by geometry.
- THE CLOSED-LOOP DOCTRINE: zero disposable stage — the jar always comes back.
- THE LINEAGE DOCTRINE: born of the Phase 162 basket — the machine died, the containers lived, and this is the container.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Tool the body: the 10/10 standard off the 162 basket geometry.
- STEP 2 — Bench the lid: spin-close seal cycles and life.
- STEP 3 — Bench the dock: plunger emptying completeness — 1G, parabolic zero-G, residual mass logged.
- STEP 4 — Run the loop: clean, refill, trade cycles; zero disposable verified.
- STEP 5 — Publish the ledger: residuals, cycle lives, the standard's drawings — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 162 — the laundry system (GP-IM-162): the parent basket; the lineage is seated law.
- PHASE 163 — the casualty pods (GP-IM-163): pod consumables ride this container.
- PHASE 159 — the agricultural matrix (GP-IM-159): harvest logistics in the same jar.
- PHASE 84 — the galley: the kitchen end of the loop.

§8 — DO-NOT-BUILD

- DO NOT pour — the plunger is the emptier; gravity is not a crew member.
- DO NOT fork the standard — one size, fleet-wide; a second size is a second logistics chain.
- DO NOT dispose — zero disposable stage is the loop's law.
- DO NOT improvise the dock — seal-lock and plunger assistance are the interface; every dock is identical.

§9 — OWED

OWED: nothing flagged — a clean green. Standing duty: residual-mass and cycle-life data publish from the bench (§6).

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-167, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, twentieth of the 20-chapter 'ok next 20, that will take us past gemini' shift (the author's order, 23 August 2026). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576 and 587): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 167